Contrastes no paramétricos

<table>
<tr><th>Parámetro</th><th>Tipo de contraste</th><th>Estadístico de contraste</th><th>Valor observado del estadístico de contraste</th><th>Contraste de hipótesis</th><th>p-valor</th></tr>
<tr><td rowspan="3">M</td><td rowspan="3">Sign</td><td rowspan="3">Si n<30: Tabla A2
$S = B\left(n, \frac{1}{2}\right)$
Si n≥30:
$S \approx N\left(\frac{n}{2}, \frac{\sqrt{n}}{2}\right)$</td><td rowspan="3">$So$= número de observaciones > m_o</td><td>$H_0: M = m_o$ vs $H_A: M > m_o$</td><td>$P\{S \geq S_o\}$</td></tr>
<tr><td>$H_0: M = m_o$ vs $H_A: M < m_o$</td><td>$P\{S \leq S_o\}$</td></tr>
<tr><td>$H_0: M = m_o$ vs $H_A: M \neq m_o$</td><td>$2 \cdot \min(P\{S \leq S_o\}, P\{S \geq S_o\})$</td></tr>
<tr><td rowspan="3">M</td><td rowspan="3">Wilcoxon signed rank</td><td rowspan="3">Si n<15: Tabla A8
Si n≥15:
$W \approx N\left(\frac{n(n+1)}{4}, \sqrt{\frac{n(n+1)(2n+1)}{24}}\right)$</td><td rowspan="3">$W_o = \sum_{i:\, X_i > m_o} R_i$</td><td>$H_0: M = m_o$ vs $H_A: M > m_o$</td><td>$P\{W \geq W_o\}$</td></tr>
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<tr><td rowspan="3">$M_{(X-Y)}$

Muestras pareadas, con mismo tamaño n</td><td rowspan="3">Wilcoxon signed rank paired</td><td rowspan="3">Si n<15: Tabla A8
Si n≥15:
$W \approx N\left(\frac{n(n+1)}{4}, \sqrt{\frac{n(n+1)(2n+1)}{24}}\right)$</td><td rowspan="3">$W_o = \sum_{i:\, D_i > 0} R_i$</td><td>$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} > 0$</td><td>$P\{W \geq W_o\}$</td></tr>
<tr><td>$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} < 0$</td><td>$P\{W \leq W_o\}$</td></tr>
<tr><td>$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} \neq 0$</td><td>$2 \cdot \min(P\{W \leq W_o\}, P\{W \geq W_o\})$</td></tr>
<tr><td rowspan="3">$M_X - M_Y$

Muestras independientes con tamaños n y m</td><td rowspan="3">Mann-Whitney-Wilcoxon rank sum</td><td rowspan="3">Si n, m <10 → Tabla A9
Si n, m ≥ 10:
$U \approx N\left(\frac{n(n+m+1)}{2}, \sqrt{\frac{n \cdot m(n+m+1)}{12}}\right)$</td><td rowspan="3">$U_o = \sum_{i:\, X_i} R_i$</td><td>$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y > 0$</td><td>$P\{U \geq U_o\}$</td></tr>
<tr><td>$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y < 0$</td><td>$P\{U \leq U_o\}$</td></tr>
<tr><td>$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y \neq 0$</td><td>$2 \cdot \min(P\{U \leq U_o\}, P\{U \geq U_o\})$</td></tr>
</table>

En las aproximaciones Normales hay que aplicar la corrección de continuidad: $P\{X < x\} = P\{X < x - 0.5\}$ ó $P\{X \leq x\} = P\{X < x + 0.5\}$